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/*
  SparkFun Inventor's Kit
  Circuit 4C - Heads Up Game

  This is a DIY version of the popular Heads Up party game. To play, one
person resets the RedBoard and holds the LCD
  facing away from them so that they cannot see it (usually on their
forehead). The display will show a short countdown
  then display random words. The other player(s) who can see the screen must
yell out clues until time runs out or the player
  guesses what word is on the screen. If they guess correctly, they can press
the button on the breadboard and another word
  will be displayed.

  This sketch was written by SparkFun Electronics, with lots of help from the
Arduino community.
  This code is completely free for any use.

  View circuit diagram and instructions at: https://learn.sparkfun.
com/tutorials/sparkfun-inventors-kit-experiment-guide---v41
  Download drawings and code at: https://github.com/sparkfun/SIK-Guide-Code
*/

#include <LiquidCrystal.h>              //the liquid crystal library contains
commands for printing to the display
LiquidCrystal lcd(13, 12, 11, 10, 9, 8);     // tell the RedBoard what pins
are connected to the display

int buttonPin = 2;                      //pin that the button is connected to
int buzzerPin = 6;                      //pin for driving the buzzer
int buttonPressTime = 0;                //variable to show how much time the
player has left to guess the word (and press the button)

long timeLimit = 15000;                 //time limit for the player to guess
each word
long startTime = 0;                     //used to measure time that has passed
for each word
int roundNumber = 0;                          //keeps track of the roundNumber
so that it can be displayed at the end of the game
const int arraySize = 25;

const char* words[arraySize] = {"moose", "beaver", "bear", "goose", "dog",
"cat", "squirrel", "bird", "elephant", "horse",
                                "bull", "giraffe", "seal", "bat", "skunk",
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"turtle", "whale", "rhino", "lion", "monkey",
                                "frog", "alligator", "kangaroo", "hippo",
"rabbit"
                              };

// the start value in the sequence array must have a value that could never
be an index of an array
// or at least a value outside the range of 0 to the size of the words array
- 1; in this case, it can't be between 0 to 24
int sequence[] = { -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,
-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1}; //start with an array full of
-1's

void setup() {

  pinMode(buttonPin, INPUT_PULLUP);        //set the button pin as an input

  lcd.begin(16, 2);                        //tell the LCD library the size of
the screen

  generateRandomOrder();                   //generate an array of random
numbers from 0 to 24 that will determine which order the words are shown in

  showStartSequence();                     //print the start sequence text

}

void loop() {

  for (int i = 0; i < arraySize; i++) {          //for each of the 25 words
in the sequence

    lcd.clear();                          //clear off the array

    roundNumber = i + 1;                    //the array starts at 0, but the
roundNumber will start counting from 1
    lcd.print(roundNumber);                 //print the roundNumber (this is
the current round number)
    lcd.print(": ");                        //spacer between the number and
the word
    lcd.print(words[sequence[i]]);         //print a random word from the word
array

    startTime = millis();  //record the time that this round started
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    while (digitalRead(buttonPin) == HIGH) {          //do this until the
button is pressed...

      int roundedTime = round((timeLimit - (millis() - startTime)) / 1000);
//calculate the time left in the round (dividing by 1000 converts the number
to seconds
      lcd.setCursor(14, 1);
//set the cursor in the lower right corner of the screen
      lcd.print("  ");
      lcd.setCursor(14, 1);
//set the cursor in the lower right corner of the screen
      lcd.print(roundedTime);
//print the time left in the time limit
      delay(15);

      if (millis() - startTime > timeLimit) {        //if the time limit is up
before the button is pressed
        gameOver();                                  //end the game
      }

      if (digitalRead(buttonPin) == LOW) {
        tone(buzzerPin, 272, 10);                    //emit a short beep when
the button is pressed
      }

    }                        //exit this loop when the button is pressed

    delay(500);              //delay for a moment before going onto the next
round, so that the button press doesn't get registered twice

  }
  //if you finish all 25 words
  winner();   //show the you win message
}


//--------------FUNCTIONS------------------------------

//DISPLAYS A COUNTDOWN TO START THE GAME
void showStartSequence() {

  lcd.clear();                  //clear the screen
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  lcd.setCursor(0, 0);           //move the cursor to the top left corner
  lcd.print("Category:");        //print "Category:"

  lcd.setCursor(0, 1);           //move the cursor to the bottom left corner
  lcd.print("Animals");          //print "Animals:"

  delay(2000);                   //Wait 2 seconds

  lcd.clear();                   //clear the screen
  lcd.print("Get ready!");       //print "Get ready!"
  delay(1000);                   //wait 1 second

  lcd.clear();                   //clear the screen
  lcd.print("3");                //print "3"
  delay(1000);                   //wait 1 second

  lcd.clear();                   //clear the screen
  lcd.print("2");                //print "3"
  delay(1000);                   //wait 1 second

  lcd.clear();                   //clear the screen
  lcd.print("1");                //print "3"
  delay(1000);                   //wait 1 second
}

//GENERATES A RANDOM ORDER FOR THE WORDS TO BE DISPLAYED
void generateRandomOrder() {

  randomSeed(analogRead(0));             //reset the random seed (Arduino
needs this to generate truly random numbers

  for (int i = 0; i < arraySize; i++) {         //do this until all 25
positions are filled

    int currentNumber = 0;               //variable to hold the current number
    boolean match = false;               //does the currentNumber match any of
the previous numbers?

    //generate random numbers until you've generated one that doesn't match
any of the other numbers in the array
    do {
      currentNumber = random(0, arraySize);             //generate a random
number from 0 to 24
      match = false;                                    //we haven't checked for
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matches yet, so start by assuming that it doesn't match
      for (int i = 0; i < arraySize; i++) {            //for all 25 numbers
in the array
        if (currentNumber == sequence[i]) {     //does the currentNumber
match any of the numbers?
          match = true;                         //if so, set the match
variable to true
        }
      }
    } while (match == true);                    //if the match variable is
true, generate another random number and try again
    sequence[i] = currentNumber;                //if the match variable is
false (the new number is unique) then add it to the sequence
  }
}

//GAME OVER
void gameOver() {
  lcd.clear();                      //clear the screen

  lcd.setCursor(0, 0);              //move the cursor the top left corner
  lcd.print("Game Over");           //print "Game Over"

  lcd.setCursor(0, 1);              //move to the bottom row
  lcd.print("Score: ");             //print a label for the score
  lcd.print(roundNumber-1);         //print the score (the score is equal to
the previous level/round number)

  //play the losing fog horn
  tone(buzzerPin, 130, 250);        //E6
  delay(275);
  tone(buzzerPin, 73, 250);         //G6
  delay(275);
  tone(buzzerPin, 65, 150);         //E7
  delay(175);
  tone(buzzerPin, 98, 500);         //C7
  delay(500);

  while (true) {}                   //get stuck in this loop forever
}

//WINNER
void winner() {
  lcd.clear();                      //clear the screen
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  lcd.setCursor(7, 0);              //move the cursor to the top center of the
screen
  lcd.print("YOU");                 //print "You"

  lcd.setCursor(7, 1);              //move the cursor to the bottom center of
the screen
  lcd.print("WIN!");                //print "WIN!"

  //play the 1Up noise
  tone(buzzerPin, 1318, 150);       //E6
  delay(175);
  tone(buzzerPin, 1567, 150);       //G6
  delay(175);
  tone(buzzerPin, 2637, 150);       //E7
  delay(175);
  tone(buzzerPin, 2093, 150);       //C7
  delay(175);
  tone(buzzerPin, 2349, 150);       //D7
  delay(175);
  tone(buzzerPin, 3135, 500);       //G7
  delay(500);

  while (true) {}                   //get stuck in this loop forever
}
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